

Finance AI Assistant — Functionalities Specification

Conversational Banking MVP + V1+ | Gemini + Supabase | Detailed behaviors

Generated on 2025-10-22 05:59:29

CONVERSATIONAL BANKING (MVP)

What it does

- Natural chat -> action JSON: { action, confidence, params, summary_human, followup_question, redteam_flag }.
- Smart follow-ups: ask a single, precise question when params are missing.
- Quick-action chips: Balance • Recent Txns • Spend Summary • Set Budget.

Inputs

- Free-form text ("balance?", "last 5 transactions", "send Rs.500 to Anjali").
- Optional quick-chip clicks.

Outputs/UX

- Assistant reply + pending Action Request card with NEEDS_CONFIRMATION when applicable.

Data & APIs

- Read/write: chat_sessions, chat_messages, parsed_intents, action_requests.

Validation & rules

- confidence < 0.85 -> ask follow-up.
- State-changing intents -> action_requests(status='NEEDS_CONFIRMATION').
- Suspected injection -> log to redteam_events.

ACCOUNTS & BALANCES (MVP)

What it does

- List user accounts with masked identifiers (e.g., XXXX-1234) and balances.
- Snapshot: available balance, holds (if modeled), last 3 activities.

Inputs

- "Show my accounts", "savings balance", "what's in credit card?"

Outputs/UX

- Expandable list (name, masked_number, currency, balance, status).

Data & APIs

- Read: accounts (or v_user_accounts), transactions (latest N).

Validation & rules

- Never show full account numbers.
- Currency formatting per currency (default INR).

TRANSACTIONS & STATEMENTS (MVP)

What it does

- History with filters (date, amount, merchant, category); keyword search.
- Exports: CSV (arbitrary ranges), monthly PDF statements.

Inputs

- "Show transactions last 30 days", "search Amazon", "download April statement".

Outputs/UX

- Paginated list; export links (CSV/PDF) in chat.

Data & APIs

- Read: transactions; write export job output links to action_requests.result.

Validation & rules

- Default date range = last 30 days; max export window (e.g., 12 months).

PAYMENTS & TRANSFERS (MOCK) (MVP)

What it does

- Own-account transfers; to saved contacts; bill pay (mock catalog).
- Confirmation step before execution; schedule one-time future transfers.

Inputs

- "Send Rs.500 to Anjali", "pay electricity bill Rs.1,200", "schedule Rs.8,000 on 5th".

Outputs/UX

- Action Request card (from, to, amount, date, note). Buttons: Confirm, Cancel, Edit.

Data & APIs

- Read: accounts, contacts. Write: action_requests, action_audit_log; mock transactions row on execute.

Validation & rules

- Validate amount/payee/balance; set executes_at for scheduled transfers.
- No real payment rails; idempotent and reversible in staging.

Security

- Require confirmation tag or Confirm button; no free-text execution.

CARDS (MVP)

What it does

- Freeze/unfreeze (mock state), view status/masked details, report lost/stolen (ticket).

Inputs

- "Freeze my card", "unfreeze travel card", "I lost my card".

Outputs/UX

- Freeze/unfreeze Action Request; ticket ID for lost/stolen.

Data & APIs

- Read/write: accounts.status (or future cards table).

Validation & rules

- Freeze needs confirmation; unfreeze requires short reason (audit log).

PFM (MVP)

What it does

- Auto-categorize spend; insights (top merchants/categories, largest expense).
- Budgets (create/view/edit; utilization %) and Goals (target amount/date).

Inputs

- "Where did I spend most?", "set Rs.5,000 food budget", "goal: Rs.50k in 3 months".

Outputs/UX

- Summary cards and progress bars.

Data & APIs

- Read: transactions aggregates; write: budgets/goals in metadata or future tables.

Validation & rules

- Enforce sane limits; warn on unrealistic inputs.

ALERTS & NOTIFICATIONS (MVP)

What it does

- Low balance, upcoming bills, unusual spend triggers.
- Alert inbox in chat; unread counter; mark as read; preferences.

Inputs

- "Show my alerts", "mute unusual spend", "mark all read".

Outputs/UX

- Alert list with quick actions.

Data & APIs

- Alerts table or action_requests logs; preferences in profiles.metadata/settings.

Validation & rules

- Backend cron/jobs; not model-driven.

PRODUCT GUIDANCE (MVP)

What it does

- Personalized offers with "Why am I seeing this?"; mock pre-qualification.

Inputs

- "Best card for me?", "eligible for premium savings?"

Outputs/UX

- Offer card with rationale; pre-qual banner (mock).

Data & APIs

- Read: spend patterns; write: dismiss/accept in action_audit_log.

Validation & rules

- Always show disclaimer; no KYC in chat.

HUMAN HANDOFF & SUPPORT (MVP)

What it does

- Escalate to human; ticket status; FAQ answers.

Inputs

- "Talk to a human", "check my ticket", "NEFT cutoff?"

Outputs/UX

- Ticket card (ID, status, last update); transcript link.

Data & APIs

- Write: action_requests(action='HUMAN_HANDOFF'); result stores ticketId.

Validation & rules

- Mask PII in transcript downloads.

PRIVACY, SECURITY & CONTROL (MVP)

What it does

- Confirmations; redaction; data export (ZIP); data deletion (mock ticket).

Inputs

- "Download my data", "delete my data".

Outputs/UX

- Download link; deletion ticket card.

Data & APIs

- Read: chat_messages/statements; write: action_requests, redteam_events.

Validation & rules

- Signed URLs with expiry; deletion is mock without admin tooling.

SETTINGS & PERSONALIZATION (MVP)

What it does

- Notification prefs; rename accounts; override categories; default account; language/theme/large text.

Inputs

- "Make checking default", "rename savings to Travel Fund", "dark mode".

Outputs/UX

- Confirmation toast and preview.

Data & APIs

- Write: profiles, accounts.name, profiles.metadata.

Validation & rules

- Owner-only updates; audit log entries.

ACCESSIBILITY (MVP)

What it does

- Keyboard navigation; screen-reader labels; high-contrast; voice input toggle + transcript.

Inputs

- "Enable voice input"; keyboard shortcuts.

Outputs/UX

- Toggles and ARIA labels.

Data & APIs

- Store prefs in profiles.metadata.

Validation & rules

- WCAG 2.1 AA; never force voice.

V1+ (Nice-to-have)

Deeper insights

- Subscription detection; cashflow forecast; "What changed this month?"; merchant insights.

Advanced payments

- New payee onboarding (IFSC/UPI checks); split transaction/bill; templates.

Card controls

- Spend limits; travel notice.

Documents & tax

- Annual spend report; tax-ready CSV by FY.

Cross-cutting guardrails

- All state changes -> NEEDS_CONFIRMATION + clear summary_human.
- Mask sensitive data; log prompt injection; throttle exports; enforce RLS across all tables.